

# API Documentation

## Overview

This document provides a comprehensive reference for all REST API endpoints in the Vidyarthi Book Depot platform. All APIs follow RESTful conventions and return JSON responses.

## Base URL

Development: `http://localhost:8080/api`

Production: `https://api.vidyarthibookdepot.com/api`

## Authentication

Most endpoints require JWT authentication. Include the token in the Authorization header:

Authorization: Bearer {jwt\_token}

## Common Response Formats

### Success Response

```
{
  "data": { ... },
  "message": "Success message",
  "timestamp": "2025-12-23T10:00:00Z"
}
```

# Error Response

```
{
  "timestamp": "2025-12-23T10:00:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Validation failed",
  "path": "/api/inquiries",
  "details": [
    {
      "field": "subject",
      "message": "Subject is required"
    }
  ]
}
```

# Paginated Response

```
{
  "content": [ ... ],
  "page": 0,
  "size": 20,
  "totalElements": 100,
  "totalPages": 5,
  "first": true,
  "last": false
}
```

# HTTP Status Codes

- 200 OK : Successful request
- 201 Created : Resource created successfully
- 204 No Content : Successful deletion
- 400 Bad Request : Invalid request data
- 401 Unauthorized : Authentication required
- 403 Forbidden : Insufficient permissions
- 404 Not Found : Resource not found
- 409 Conflict : Resource conflict (e.g., duplicate)
- 500 Internal Server Error : Server error

# API Endpoints

## Authentication APIs

### Register User

POST /api/auth/register

Content-Type: application/json

```
{  
  "email": "customer@example.com",  
  "phone": "+919876543210",  
  "password": "SecurePassword123!",  
  "firstName": "John",  
  "lastName": "Doe"  
}
```

**Response:** 201 Created

### Login

POST /api/auth/login

Content-Type: application/json

```
{  
  "email": "customer@example.com",  
  "password": "SecurePassword123!"  
}
```

**Response:** 200 OK

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "type": "Bearer",
  "expiresIn": 3600,
  "user": {
    "id": 1,
    "email": "customer@example.com",
    "firstName": "John",
    "lastName": "Doe",
    "role": "CUSTOMER"
  }
}
```

## Refresh Token

POST /api/auth/refresh

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{
  "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

**Response:** 200 OK

## User APIs

### Get Current User Profile

GET /api/users/profile

**Authorization:** Bearer {token}

**Response:** 200 OK

## Update User Profile

PUT /api/users/profile

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "phone": "+919876543211",  
  "firstName": "Jane",  
  "lastName": "Smith"  
}
```

**Response:** 200 OK

## Get User by ID (Owner Only)

GET /api/users/{id}

**Authorization:** Bearer {token}

**Response:** 200 OK

## Update User Role (Owner Only)

PUT /api/users/{id}/role

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "role": "OWNER"  
}
```

**Response:** 200 OK

## List All Users (Owner Only)

GET /api/users?page=0&size=20&role=CUSTOMER&enabled=true

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

# Product APIs

## List Products

GET /api/products?page=0&size=20&categoryId=1&board=SSC&class=10&search=mathematics

**Response:** 200 OK (Paginated)

## Get Product Details

GET /api/products/{id}

**Response:** 200 OK

## Search Products

GET /api/products/search?q=mathematics&page=0&size=20

**Response:** 200 OK (Paginated)

## Get Products by Category

GET /api/products/category/{categoryId}?page=0&size=20

**Response:** 200 OK (Paginated)

## Get Products by Board

GET /api/products/board/{board}?page=0&size=20

**Response:** 200 OK (Paginated)

## Get Products by Class

GET /api/products/class/{class}?page=0&size=20

**Response:** 200 OK (Paginated)

## Create Product (Owner Only)

POST /api/products

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{
  "name": "Physics for Class 12",
  "description": "Advanced physics textbook",
  "isbn": "978-1234567891",
  "author": "Dr. Suresh Patel",
  "publisher": "Science Publishers",
  "price": 550.00,
  "categoryId": 1,
  "board": "HSC",
  "class": "12",
  "featured": false,
  "bestseller": true,
  "imageUrl": "/images/products/physics-12.jpg"
}
```

**Response:** 201 Created

## Update Product (Owner Only)

PUT /api/products/{id}

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{
  "name": "Updated Product Name",
  "price": 600.00
}
```

**Response:** 200 OK

## Delete Product (Owner Only)

DELETE /api/products/{id}

**Authorization:** Bearer {token}

**Response:** 204 No Content

## Get Featured Products

GET /api/products/featured?limit=10

**Response:** 200 OK

## Get Bestseller Products

GET /api/products/bestsellers?limit=10

**Response:** 200 OK

## Get Trending Products

GET /api/products/trending?limit=10&period=MONTH

**Response:** 200 OK

## Category APIs

### List All Categories

GET /api/categories?includeSubcategories=false

**Response:** 200 OK

### Get Category by ID

GET /api/categories/{id}

**Response:** 200 OK

### Get Category Tree

GET /api/categories/tree

**Response:** 200 OK



## Create Category (Owner Only)

POST /api/categories

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "name": "Competitive Exams",  
  "description": "Books for competitive exam preparation",  
  "parentCategoryId": null  
}
```

**Response:** 201 Created

## Update Category (Owner Only)

PUT /api/categories/{id}

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "name": "Updated Category Name",  
  "description": "Updated description"  
}
```

**Response:** 200 OK

## Delete Category (Owner Only)

DELETE /api/categories/{id}

**Authorization:** Bearer {token}

**Response:** 204 No Content

# Inquiry APIs

## Create Inquiry

POST /api/inquiries

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{
  "type": "BOOK_AVAILABILITY",
  "subject": "Availability of Mathematics Class 10",
  "message": "Do you have Mathematics for Class 10 in stock?",
  "productId": 1
}
```

**Response:** 201 Created

## List Inquiries

GET /api/inquiries?page=0&size=20&status=PENDING&type=BOOK\_AVAILABILITY

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

## Get Inquiry Details

GET /api/inquiries/{id}

**Authorization:** Bearer {token}

**Response:** 200 OK

## Update Inquiry Status (Owner Only)

PUT /api/inquiries/{id}/status

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{
  "status": "RESOLVED"
}
```

**Response:** 200 OK

## Add Response to Inquiry (Owner Only)

POST /api/inquiries/{id}/response

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "message": "Yes, we have this book in stock."  
}
```

**Response:** 201 Created

## Get My Inquiries

GET /api/inquiries/my-inquiries?page=0&size=20&status=PENDING

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

## Get Pending Inquiries (Owner Only)

GET /api/inquiries/pending?page=0&size=20

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

## Get Inquiry Statistics (Owner Only)

GET /api/inquiries/statistics?period=MONTH

**Authorization:** Bearer {token}

**Response:** 200 OK

## Update Inquiry

```
PUT /api/inquiries/{id}
Authorization: Bearer {token}
Content-Type: application/json

{
  "subject": "Updated subject",
  "message": "Updated message"
}
```

**Response:** 200 OK

## Delete Inquiry

```
DELETE /api/inquiries/{id}
Authorization: Bearer {token}
```

**Response:** 204 No Content

## Chat APIs

### Create or Get Conversation

```
POST /api/chat/conversations
Authorization: Bearer {token}
Content-Type: application/json

{}
```

**Response:** 200 OK or 201 Created

### List User's Conversations

```
GET /api/chat/conversations?page=0&size=20
Authorization: Bearer {token}
```

**Response:** 200 OK (Paginated)

## Get Conversation with Messages

GET /api/chat/conversations/{id}?page=0&size=50

**Authorization:** Bearer {token}

**Response:** 200 OK

## Send Message

POST /api/chat/conversations/{id}/messages

**Authorization:** Bearer {token}

**Content-Type:** application/json

```
{  
  "message": "Hello, do you have Mathematics Class 10?"  
}
```

**Response:** 201 Created

## Get Messages

GET /api/chat/conversations/{id}/messages?page=0&size=50

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

## Mark Message as Read

PUT /api/chat/messages/{id}/read

**Authorization:** Bearer {token}

**Response:** 200 OK

## Get Unread Message Count

GET /api/chat/unread-count

**Authorization:** Bearer {token}

**Response:** 200 OK

## Get All Conversations (Owner Only)

GET /api/chat/conversations/owner/all?page=0&size=20&unreadOnly=false

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

## Notification APIs

### Get User Notifications

GET /api/notifications?page=0&size=20&read=false&type=INQUIRY\_RESPONSE

**Authorization:** Bearer {token}

**Response:** 200 OK (Paginated)

### Mark Notification as Read

PUT /api/notifications/{id}/read

**Authorization:** Bearer {token}

**Response:** 200 OK

### Mark All Notifications as Read

PUT /api/notifications/read-all

**Authorization:** Bearer {token}

**Response:** 200 OK

### Get Unread Notification Count

GET /api/notifications/unread-count

**Authorization:** Bearer {token}

**Response:** 200 OK

# WebSocket API

## Connection

```
const socket = new SockJS('http://localhost:8080/ws/chat');
const stompClient = Stomp.over(socket);

stompClient.connect({}, function(frame) {
    console.log('Connected: ' + frame);

    // Subscribe to conversation messages
    stompClient.subscribe('/topic/chat/1', function(message) {
        const chatMessage = JSON.parse(message.body);
        console.log('Received:', chatMessage);
    });
});
```

## Send Message via WebSocket

```
stompClient.send("/app/chat/send", {}, JSON.stringify({
    conversationId: 1,
    message: "Hello, do you have Mathematics Class 10?"
}));
```

## Rate Limiting

API endpoints are rate-limited to prevent abuse:

- **Authentication endpoints:** 5 requests per minute per IP
- **General endpoints:** 100 requests per minute per user
- **Chat endpoints:** 30 messages per minute per user

Rate limit headers:

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 95
X-RateLimit-Reset: 1703328600
```

# Error Handling

## Validation Errors

When request validation fails:

```
{
  "timestamp": "2025-12-23T10:00:00Z",
  "status": 400,
  "error": "Validation Failed",
  "message": "Invalid request data",
  "details": [
    {
      "field": "email",
      "message": "Email is required"
    },
    {
      "field": "password",
      "message": "Password must be at least 8 characters"
    }
  ]
}
```

## Authentication Errors

```
{
  "timestamp": "2025-12-23T10:00:00Z",
  "status": 401,
  "error": "Unauthorized",
  "message": "Invalid or expired token"
}
```



## Authorization Errors

```
{  
  "timestamp": "2025-12-23T10:00:00Z",  
  "status": 403,  
  "error": "Forbidden",  
  "message": "You do not have permission to access this resource"  
}
```

## Not Found Errors

```
{  
  "timestamp": "2025-12-23T10:00:00Z",  
  "status": 404,  
  "error": "Not Found",  
  "message": "Resource not found",  
  "path": "/api/products/999"  
}
```

## API Versioning

Currently, all APIs are unversioned. Future versions will use:

```
/api/v1/products  
/api/v2/products
```

## CORS Configuration

CORS is enabled for the following origins:

- Development: <http://localhost:3000>
- Production: <https://vidyarthibookdepot.com>

# Request/Response Examples

## Complete Product Creation Flow

# 1. Login as owner

```
curl -X POST http://localhost:8080/api/auth/login \  
  -H "Content-Type: application/json" \  
  -d '{  
    "email": "owner@vbd.com",  
    "password": "password123"  
  }'
```

# Response contains JWT token

```
TOKEN="eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
```

# 2. Create product

```
curl -X POST http://localhost:8080/api/products \  
  -H "Authorization: Bearer $TOKEN" \  
  -H "Content-Type: application/json" \  
  -d '{  
    "name": "Mathematics for Class 10",  
    "description": "Comprehensive mathematics textbook",  
    "isbn": "978-1234567890",  
    "author": "Dr. Ramesh Kumar",  
    "publisher": "Educational Publishers",  
    "price": 450.00,  
    "categoryId": 1,  
    "board": "SSC",  
    "class": "10",  
    "featured": true  
  }'
```

# Complete Inquiry Flow

# 1. Customer creates inquiry

```
curl -X POST http://localhost:8080/api/inquiries \  
  -H "Authorization: Bearer $CUSTOMER_TOKEN" \  
  -H "Content-Type: application/json" \  
  -d '{  
    "type": "BOOK_AVAILABILITY",  
    "subject": "Availability of Mathematics Class 10",  
    "message": "Do you have this book in stock?",  
    "productId": 1  
  }'
```

# 2. Owner responds

```
curl -X POST http://localhost:8080/api/inquiries/1/response \  
  -H "Authorization: Bearer $OWNER_TOKEN" \  
  -H "Content-Type: application/json" \  
  -d '{  
    "message": "Yes, we have this book in stock."  
  }'
```

## Testing

### Postman Collection

A Postman collection is available with all API endpoints pre-configured.

### API Testing Tools

- Postman
- cURL
- HTTPie
- Swagger UI (available at `/swagger-ui.html` )

## Support

For API support, contact:

- Email: [support@vidyarthibookdepot.com](mailto:support@vidyarthibookdepot.com)

- Documentation: <https://docs.vidyarthibookdepot.com>